ELEC60008
ELEC70089

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

EEE/EIE PART III/IV: MEng, BEng and ACGI

CONTROL ENGINEERING

Monday, 11 December 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Questions ONE, TWO and THREE carry 40, 30 and 30 marks respectively.

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following short questions accounting, overall, for 40% of the exam mark (that is 40 marks out of 100).

a) Let $A \in \mathbb{R}^{n\times n}$ and assume $\det(A) \neq 0$. Let

$$p(\lambda) = \det(\lambda I - A) = \lambda^n + \alpha_{n-1}\lambda^{n-1} + \ldots + \alpha_1\lambda + \alpha_0$$

be its characteristic polynomial.

Using Cayley-Hamilton theorem determine an expression for A^{-n} in terms of the coefficients α_i, $i = 0,\ldots,n-1$, and $I_n, A^{-1}, A^{-2}\ldots,A^{-(n-1)}$ where I_n is the $n \times n$ identity matrix. [4]

b) Consider the bidimensional nonlinear system of equations

$$\begin{aligned} \dot{x}_1 &= x_1(1-x_1) - x_1x_2 \\ \dot{x}_2 &= -x_2 + 2x_1x_2 \end{aligned}$$

describing the evolutions of two interacting populations of animals (preys and predators, respectively).

i) Find all equilibrium points of the system. [4]

ii) Write the equation of the linearized system around each equilibrium previously computed. [3]

iii) Discuss if it is possible to use the principle of stability in the first approximation and, if this is so, study the stability properties of each equilibrium. [5]

c) Consider the continuous-time linear system of equations:

$$\dot{x} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u.$$

Find the values of scalars α and β such that a piece-wise constant input u of the form:

$$u(t) = \begin{cases} \alpha & t \in [0,1] \\ \beta & t \in [1,2] \end{cases}$$

transfers the initial state $x(0) = [0,0]^T$ to the final state $x(2) = [3,0]^T$. (*Hint: recall the formula* $x(t) = e^{At}x(0) + \int_0^t e^{A(t-s)}Bu(s)\,ds$). [6]

Show that it is not possible to define $u(t)$ for $t \geq 2$ such that $x(t) = [3,0]^T$ for all subsequent times. [2]

d) Consider the discrete time system:

$$x_1^+ = x_1 + \alpha x_2 \qquad x_2^+ = -x_2 + u \qquad y = x_1 + x_2$$

where $\alpha \in \mathbb{R}$ is a constant parameter, u is a scalar input and y is the scalar output.

i) Write the system in matrix representation. [2]

ii) Find under what conditions on α the system is reachable [2]

iii) Find under what conditions on α the system is observable [2]

iv) Find under what conditions on α the system is controllable [2]

e) Consider the 3 dimensional discrete time system:

$$x^+ = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} x, \qquad y = [1,0,0]x$$

i) Show that the system is observable. [4]

ii) Find an initial condition $x(0)$ such that the above system's output solution fulfills the periodicity condition:

$$y(0), y(1), \ldots, y(5), y(6), \ldots = a, b, c, -a, -b, -c, a, b, c, -a, -b, -c, \ldots$$

where a, b and c are arbitrary constants. [4]

2. Consider two rotating masses, of inertia J_1 and J_2 respectively, elastically coupled (through a rotational elastic constant) k, and subject to the action of a torque u, which is acting on the first mass. Assume both masses are subject to a dynamic friction (of coefficient β) proportional to their angular speed. This set-up can be modeled according to a system of second-order differential equations, as follows:

$$J_1\ddot{\theta}_1 = -k(\theta_1 - \theta_2) + u - \beta\dot{\theta}_1$$

$$J_2\ddot{\theta}_2 = -k(\theta_2 - \theta_1) - \beta\dot{\theta}_2$$

Assume that a sensor measures the angular position of the second mass θ_2.

a) Assume $x_1 = \theta_1$, $x_2 = \dot{\theta}_1$, $x_3 = \theta_2$ and $x_4 = \dot{\theta}_2$. Under these assignments, write a state-space representation of the rotating mass system outlined above, that is in the form

$$\dot{x} = Ax + Bu, \qquad y = Cx$$

for suitable matrices A, B and C. [8]

For the rest of the question assume, for simplicity, $J_1 = J_2 = 1$, $k = 1$ and $\beta = 1$.

b) Show that the system previously determined is controllable. [6]

c) Show that the system is observable. [6]

d) Verify that the inverse of the observability matrix is given by

$$\mathscr{O}^{-1} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

[2]

Design a Luenberger observer for the system so as to place all closed-loop poles at -1. (Sketch the procedure through formulas if computations are too long)

[8]

3. Consider the linear time-invariant system:

$$\dot{x}(t) = Ax(t)$$

with $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$, such that all of its eigenvalues have negative real part. Consider an arbitrary symmetric positive definite matrix $Q \in \mathbb{R}^{n\times n}$.

a) Show that the matrix-valued function $\eta(t) := e^{A^T t} Q e^{At}$, is symmetric, for all $t \in \mathbb{R}$ [3]

b) Show that $\eta(t)$ is a solution of the following matrix-valued ODE:

$$\dot{X}(t) = A^T X(t) + X(t)A, \qquad X(0) = Q$$

[5]

c) Define the matrix P as follows:

$$P = \int_0^{+\infty} \eta(t)\, dt.$$

Show that $P \in \mathbb{R}^{n\times n}$ is well-defined (i.e. finite), and symmetric. [5]

d) A generic matrix P is positive definite if and only if $x_0^T P x_0 > 0$ for all non-zero $x_0 \in \mathbb{R}^n$. Show, by applying this definition, that the matrix P of part (c) is positive definite (*Hint: bring the vector x_0 inside the integral defining P.*) [5]

e) Exploiting the identity $A^T \eta(t) + \eta(t)A = \dot{\eta}(t)$, prove for the matrix P of part (c) the following:

$$A^T P + PA = -Q$$

(*Hint: compute $A^T P + PA$ while using the integral definition of P. Bring A and A^T within the integral and apply the fundamental theorem of calculus.*) [5]

f) Assume:

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}$$

and $Q = I_2$, the identity matrix of dimension 2. Explain why a positive definite and symmetric solution P of the following equation exists:

$$A^T P + PA = -Q$$

[2]

Compute P, the solution of the above equation for the values of A and Q given above. [5]